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Linear Algebra

Vector spaces

$\mathbb{R}^n = \{ (x_1, x_2, \ldots, x_n) : x_1, \ldots, x_n \in \mathbb{R}^n \}$

→ Addition:

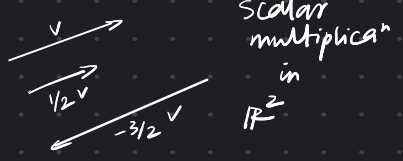
$x, y \in \mathbb{R}^n \qquad x+y = (x_1, x_2, \ldots, x_n) + (y_1, y_2, \ldots, y_n)$

$= (x_1+y_1, x_2+y_2, \ldots, x_n+y_n) \in \mathbb{R}^n$ → closure of addition

→ Scalar multiplicaⁿ:

$\lambda \in \mathbb{R}, \ x \in \mathbb{R}^n \qquad \lambda x = \lambda(x_1, \ldots, x_n)$

$= (\lambda x_1, \ldots, \lambda x_n) \in \mathbb{R}^n$



Vector space

A vector space is a set V with 2 operations defined on it: vector addition and scalar multiplication, such that V follows the following 8 properties.

let's consider $V = \mathbb{R}^n$ on field $\mathbb{R}$

1. Closure of addition:

   If $x, y \in \mathbb{R}^n$ then $x+y \in \mathbb{R}^n$

2. Closure of multiplication:

   If $x \in \mathbb{R}^n, \lambda \in \mathbb{R}$, then $\lambda x \in \mathbb{R}^n$

3. Commutativity of addition:

   $x, y \in \mathbb{R}^n \Rightarrow x+y = y+x$

4. Associativity:

   $w, x, y \in \mathbb{R}^n \qquad \lambda_1, \lambda_2 \in \mathbb{R}$

   For vector addition

   $(w+x)+y = w+(x+y)$

   For scalar multiplication

   $(\lambda_1 \lambda_2) x = \lambda_1(\lambda_2 x) = \lambda_2(\lambda_1 x)$

5. Additive identity:

   There exists a zero vector.

   $0 \in \mathbb{R}^n$ s.t. $x+0 = x$

6. Additive inverse:

   $\forall x \in \mathbb{R}^n$,

   $\exists y \in \mathbb{R}^n$ s.t. $x+y = 0$ → additive identity

   $y = -x$ → additive inverse

7. Multiplicative identity:

   $\forall x \in \mathbb{R}^n$,

   $\exists 1$ s.t. $1 \cdot x = x$

8. Distributive properties:

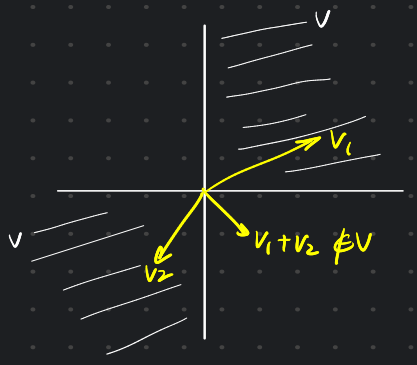
$x, y \in \mathbb{R}^n$
$a, b \in \mathbb{R}$

$$a(x+y) = ax + ay$$

$$(a+b)x = ax + bx$$

Q. $V = \{(x, y) : xy \geq 0,\ x, y \in \mathbb{R}\}$. Is this a vector space?

→ possible op^{n}s: $x, y > 0$, $x, y < 0$, $x = 0$, $y = 0$ } cases where $(x, y) \in V$



① closed under addition:

$(x_1, y_1), (x_2, y_2) \in V$

consider an example: $(2, 3)$ and $(-3, -2)$

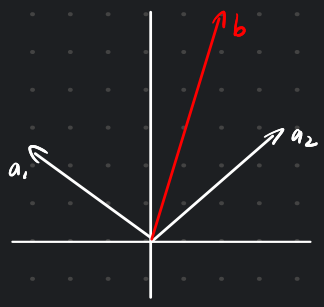
$(2, 3) + (-3, -2) = (-1, 1) \notin V$ as $(-1 \times 1) = -1 < 0$

NOT a vector space.

Systems of linear equations

Say we have 2 vectors a1 and a2. We need to represent a third vector b as a linear combination of a1 and a2.

$$a_1 = \begin{bmatrix} -2 \\ 2 \end{bmatrix} \quad a_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$



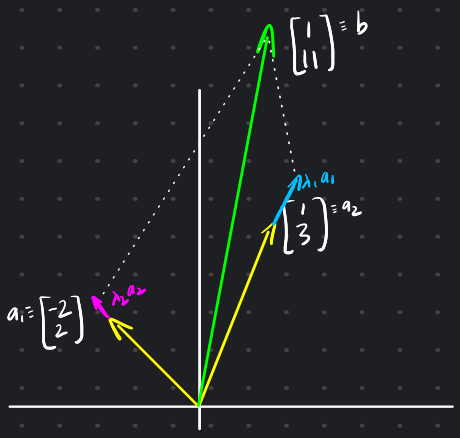
Essentially we need λ1 and λ2 s.t.

$$\lambda_1 a_1 + \lambda_2 a_2 = b$$

$$\lambda_1 \begin{bmatrix} -2 \\ 2 \end{bmatrix} + \lambda_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$\Rightarrow \quad -2\lambda_1 + \lambda_2 = 1$$

$$2\lambda_1 + 3\lambda_2 = 11$$



Matrix form:

$$\begin{bmatrix} -2 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$\Rightarrow \left(\lambda_1 \begin{bmatrix} -2 \\ 2 \end{bmatrix} + \lambda_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 11 \end{bmatrix} \quad \Rightarrow \begin{bmatrix} -2\lambda_1 + \lambda_2 \\ -2\lambda_2 + 3\lambda_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

Essentially we'll need to solve this system of linear equations.

Example of matrix vector multiplication

$$\begin{pmatrix} 1 & 2 & 4 \\ -2 & 3 & 1 \\ -4 & 1 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix} = 2 \begin{bmatrix} 1 \\ -2 \\ -4 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \\ 8 \end{bmatrix} + \begin{bmatrix} 4 \\ 6 \\ 2 \end{bmatrix} + \begin{bmatrix} 12 \\ 3 \\ 6 \end{bmatrix} = \begin{bmatrix} 18 \\ 5 \\ 0 \end{bmatrix}$$

vectors

Matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & \cdots & \cdots & \cdots \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}_{m \times n}$$

Vector addition and scalar multiplication can be done on matrices.

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{bmatrix} = \begin{bmatrix} (a_{11}+b_{11}) & (a_{12}+b_{12}) & \cdots & (a_{1n}+b_{1n}) \\ \vdots & \vdots & \ddots & \vdots \\ (a_{m1}+b_{m1}) & (a_{m2}+b_{m2}) & \cdots & (a_{mn}+b_{mn}) \end{bmatrix}$$

$$\lambda \cdot \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} \lambda a_{11} & \lambda a_{12} & \cdots & \lambda a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda a_{m1} & \lambda a_{m2} & \cdots & \lambda a_{mn} \end{bmatrix}$$

Scalar multiplication shrinks or stretches the matrices.

Matrix multiplication

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}_{2\times3} \qquad B = \begin{bmatrix} 3 & 6 \\ 4 & 7 \\ 5 & 8 \end{bmatrix}_{3\times2}$$

(b_1, b_2)

$$A \times B = \begin{bmatrix} Ab_1 & Ab_2 \end{bmatrix}$$

$$= \begin{bmatrix} 3\begin{bmatrix}1\\2\end{bmatrix} + 4\begin{bmatrix}2\\2\end{bmatrix} + 5\begin{bmatrix}1\\3\end{bmatrix} & 6\begin{bmatrix}1\\2\end{bmatrix} + 7\begin{bmatrix}2\\2\end{bmatrix} + 8\begin{bmatrix}1\\3\end{bmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} 1\cdot3+2\cdot4+1\cdot5 & 1\cdot6+2\cdot7+1\cdot8 \\ 2\cdot3+2\cdot4+3\cdot5 & 2\cdot6+2\cdot7+3\cdot8 \end{bmatrix} = \begin{bmatrix} 16 & 28 \\ 26 & 44 \end{bmatrix}_{2\times2}$$

$$A_{m\times n} \times B_{n\times p} = C_{m\times p}$$

Laws of matrices

Addition:

1. $A+B=B+A$ ··· commutativity
2. $c(A+B)=cA+cB$ ··· distributive
3. $A+(B+C)=(A+B)+C$ ··· associativity

Multiplication:

1. $C(A+B)=CA+CB$ ··· distributive from left
2. $(A+B)C=AC+BC$ ··· distributive from right
3. $A(BC)=(AB)C$ ··· associativity

↗ could be equal in some cases, but not necessarily.

NOTE: $AB \neq BA$ [The vector space changes]

Laws of exponents over square matrices:

1. $A^n = A A \cdots A$ (↳ n times)
2. $A^n A^q = A^{n+q}$
3. $(A^n)^q = A^{n \cdot q}$

Defining $\mathbb{R}^n$:

Set of all lists/ordered set of length n where the list is made up of real #s.

$$\mathbb{R}^n = \{(x_1, x_2, \ldots, x_n) : x_1, \ldots, x_n \in \mathbb{R}\}$$

not a vector space until we define vector addition and scalar multiplication

Addition in $\mathbb{R}^n$:

$x, y \in \mathbb{R}^n \Rightarrow x = (x_1, \ldots, x_n),\ y = (y_1, \ldots, y_n)$

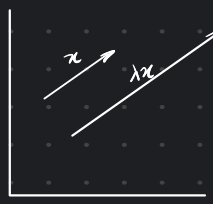
$x + y = (x_1 + y_1,\ x_2 + y_2,\ \ldots,\ x_n + y_n) \in \mathbb{R}^n$

Scalar multiplication in $\mathbb{R}^n$:

$x \in \mathbb{R}^n,\ x = (x_1, \ldots, x_n)$

$\lambda \in \mathbb{R}$

$$\lambda x = \lambda(x_1, \ldots, x_n) = (\lambda x_1, \ldots, \lambda x_n) \in \mathbb{R}^n$$



If $\lambda > 1$, λx is stretched.

$\lambda \in (0, 1)$ λx is shrunk.

Commutativity of addition in $\mathbb{R}^n$:

If $x, y \in \mathbb{R}^n$, then $x + y = y + x$

Proof: $(x+y) = (x_1 + y_1, \ldots, x_n + y_n)$

$= (y_1 + x_1, \ldots, y_n + x_n)$ .. real #s are commutative.

$= (y + x)$.

Additive inverse in $\mathbb{R}^n$:

For $x \in \mathbb{R}^n$, the additive inverse of x $(-x)$ is also a vector in $\mathbb{R}^n$ s.t.

$$x + (-x) = 0 \in \mathbb{R}^n$$

↳ zero vector $(0, 0, \ldots, 0)$

Vector space

A vector space is a set V, having the vector addition and scalar multiplication defined for it, and following the following properties:

Closure under addition:

$x + y \in V$, if $x, y \in V$

Closure under scalar multiplication:

$\lambda x \in V$, if $x \in V,\ \lambda \in \mathbb{F}$

Commutativity of addition:

$x + y = y + x$, if $x, y \in V$

Associativity of addition and scalar multiplication:

$(u + v) w = u + (v + w)$

$(ab) v = a(bv)$ $\quad u, v, w \in V,\ a, b \in \mathbb{F}$

Existence of additive identity:

$\exists\ 0 \in V$ s.t. $v + 0 = v \quad \forall v \in V$ $\quad$ e.g. $(0, 0, \ldots, 0)$ in $\mathbb{R}^n$

Existence of additive inverse:

$\forall v \in V,\ \exists w \in V$

s.t. $v + w = 0 \in V$

Existence of multiplicative identity:

$\exists\ 1 \in \mathbb{F}$ s.t.

$1 \cdot v = v \quad \forall v \in V$

Binary modulo vector space

$Z = \{0, 1\}$ $\quad x+y = (x+y) \% 2$

$\lambda \cdot x = (\lambda x) \% 2$

To check if it is a vector space:

① closed under addition:

$0+0 = (0+0)\%2 = 0 \in Z$

$0+1 = (0+1)\%2 = 1 \in Z$

$1+0 = (1+0)\%2 = 1 \in Z$

$1+1 = (1+1)\%2 = 0 \in Z$

∴ The space is closed under addition. ✓

② closed under scalar multiplicⁿ:

$\lambda \in \mathbb{R}$, $x \in Z$.

$\lambda x = \lambda x \% 2$

= either $0 \% 2$ (↓ 0) or $\lambda \% 2$ (↓ 0 or 1) $\in Z$

✓

③ commutativity of addition:

$x, y \in Z$

$x+y = (x+y) \% 2$

$= (y+x) \% 2$

$= y+x$

✓

④ ⑧ follow as well.

⑤ Additive identity: (existence of $\vec{0}$)

$x \in Z$

$x+0 = (x+0) \% 2$

$= x \% 2$

$= x$ [x is either 0 or 1, so $0\%2 = 0$, $1\%2 = 1$]

$\in Z$

⑥ Additive inverse:

$\exists w \ (w+x = 0) \ \forall x \in Z$

$x + w = 0$

$x + w = (x+w) \% 2$

If $x = 0$, $(0+w)\%2 = 0$, $\therefore w = 0 \in Z$

If $x = 1$, $(1+w)\%2 = 0$, $w = 1 \in Z$

x itself is the additive inverse of x.

⑦ multiplicative Identity:

$1x = 1x \% 2$

$= x$

$1x = x \quad \forall x \in Z$

✓

It is a vector space.

Q. $\mathbb{F}^S$: set of func"s from $S \to \mathbb{F}$.

For $f, g \in \mathbb{F}^S$, $f+g$ is defined as

$$(f+g)(x) = f(x) + g(x) \quad x \in S$$

For $\lambda \in \mathbb{F}$ and $f \in \mathbb{F}^S$, $\lambda f \in \mathbb{F}^S$ s.t.

$$(\lambda f)(x) = \lambda \cdot f(x) \quad x \in S.$$

→ To check if $\mathbb{F}^S$ is a vector space :

① closed under addition :

$$(f+g)(x) = f(x) + g(x) \in \mathbb{F}$$

② closed under multiplicat" :

$$(\lambda f)(x) = \lambda \cdot f(x) \in \mathbb{F}$$

⑤ Additive identity :

$$(f+0)(x) = f(x) + 0 = f(x) \in \mathbb{F}$$

⋮

Derived properties of vector spaces

1. A vector space has a unique additive identity.

   0 is unique.

2. A vector space has a unique additive inverse.

   w is unique for any $x \in V$ s.t. $x + w = 0$

3. $0 \cdot v = v$ (scalar, vector)

4. $\lambda \cdot 0 = 0$

5.

Subspaces

A subset U of a vector space V is called a subspace of V if U is also a vector space under the same addition and scalar multiplication operations defined on V.

Subspaces of R^2

1. $\mathbb{R}^2$ itself
2. $\vec{0} = \{(0,0)\}$
3. All lines that pass through the origin $(\mathbb{R}^1)$.

{0} and V are always subspaces of vector V.

To check if U is a subspace:
option 1: check if U is a vector space on the same F
option 2: check only closure under addition and scalar multiplication and existence of 0.

Subspaces of R^3

1. $\vec{0}$ — dim 0
2. Lines passing through origin — dim 1
3. Planes passing through origin — dim 2
4. $\mathbb{R}^3$ itself — dim 3

Q.47	Select all choices that are subspaces of $\mathbb{R}^3$. **Note:** $\mathbb{R}$ denotes the set of real numbers.
(A) ✓	$\left\{\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : \mathbf{x} = \alpha \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \alpha, \beta \in \mathbb{R}\right\}$
(B)	$\left\{\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : \mathbf{x} = \alpha^2 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + \beta^2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \alpha, \beta \in \mathbb{R}\right\}$
(C) ✓	$\left\{\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : 5x_1 + 2x_3 = 0, 4x_1 - 2x_2 + 3x_3 = 0\right\}$
(D)	$\left\{\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : 5x_1 + 2x_3 + 4 = 0\right\}$

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(A) → ① existence of $\vec{0}$: yes. $\alpha = \beta = 0$

② closure under addition:

$$\begin{bmatrix} \alpha_1 \\ \alpha_1 \\ 0 \end{bmatrix} + \begin{bmatrix} \beta_1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} \alpha_2 \\ \alpha_2 \\ 0 \end{bmatrix} + \begin{bmatrix} \beta_2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \alpha_1+\alpha_2 \\ \alpha_1+\alpha_2 \\ 0 \end{bmatrix} + \begin{bmatrix} \beta_1+\beta_2 \\ 0 \\ 0 \end{bmatrix} \in V$$

③ closed under scalar multiplication:

$$\begin{bmatrix} \lambda\alpha_1 \\ \lambda\alpha_2 \\ 0 \end{bmatrix} + \begin{bmatrix} \lambda\beta_1 \\ 0 \\ 0 \end{bmatrix} \in V$$

∴ It is a subspace.

(B) → ① zero exists: $\alpha = \beta = 0$ ✓

②

$$\begin{bmatrix} \alpha_1^2 \\ 2\alpha_1^2 \\ 0 \end{bmatrix} + \begin{bmatrix} \beta_1^2 \\ 0 \\ \beta_1^2 \end{bmatrix} + \begin{bmatrix} \alpha_2^2 \\ 2\alpha_2^2 \\ 0 \end{bmatrix} + \begin{bmatrix} \beta_2^2 \\ 0 \\ \beta_2^2 \end{bmatrix} = \begin{bmatrix} \alpha_1^2+\alpha_2^2 \\ 2\alpha_1^2+2\alpha_2^2 \\ 0 \end{bmatrix} + \begin{bmatrix} \beta_1^2+\beta_2^2 \\ 0 \\ \beta_1^2+\beta_2^2 \end{bmatrix}$$

$\notin V$

Not a subspace.

③ $\lambda x = \begin{bmatrix} \lambda\alpha^2 \\ 2\lambda\alpha^2 \\ 0 \end{bmatrix} + \begin{bmatrix} \lambda\beta^2 \\ 0 \\ \lambda\beta^2 \end{bmatrix}$ should be in form of $\begin{bmatrix} (\alpha')^2 \\ 2(\alpha')^2 \\ 0 \end{bmatrix} + \begin{bmatrix} (\beta')^2 \\ 0 \\ (\beta')^2 \end{bmatrix}$ for some $\alpha', \beta' \in \mathbb{R}$

$\alpha', \beta' \in \mathbb{R}$ s.t. $\lambda\alpha^2 = (\alpha')^2$ and $\lambda\beta^2 = (\beta')^2$ $\forall \lambda \in \mathbb{R}$

$\lambda\beta^2 = (\beta')^2$

$\beta' = \sqrt{\lambda}\,\beta$ $\forall \lambda \in \mathbb{R}$.

Invalid for $\lambda = (-1)$ as $\sqrt{-1}\,\beta \notin \mathbb{R}$.

NOT closed under scalar multiplication.

(C) → ① existence of $\vec{0}$: $x_1 = x_2 = x_3 = 0$ ✓

② closure under addition:

$$\frac{5x_1 + 2x_3 = 0 \qquad 5y_1 + 2y_3 = 0}{5(x_1+y_1) + 2(x_3+y_3) = 0}$$

$$\frac{4x_1 - 2x_2 + 3x_3 = 0 \qquad 4y_1 - 2y_2 + 3y_3 = 0}{4(x_1+y_1) - 2(x_2+y_2) + 3(x_3+y_3) = 0}$$

$\therefore \begin{bmatrix} x_1+y_1 \\ x_2+y_2 \\ x_3+y_3 \end{bmatrix} \in V$ ✓

③ closure under scalar multiplication:

$\begin{bmatrix} \lambda x_1 \\ \lambda x_2 \\ \lambda x_3 \end{bmatrix} \in V$?

$5(\lambda x_1) + 2(\lambda x_3) = \lambda[5x_1 + 2x_3] = \lambda \cdot 0 = 0$

$4(\lambda x_1) - 2(\lambda x_2) + 3(\lambda x_3) = \lambda[4x_1 - 2x_2 + 3x_3] = \lambda \cdot 0 = 0$

$\lambda\vec{x} \in V$ ✓✓

It is a subspace.

(D) → ① existence of $\vec{0}$:

$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ $5(0) + 2(0) + 4 = 4 \neq 0$

NOT a subspace.

H.W 1 (Prove!) If $b \in \mathbb{R}$, then

$$U = \{(x_1, x_2, x_3, x_4) \in \mathbb{R}^4 : x_3 = 5x_4 + b\}$$

is a subspace iff $b = 0$.

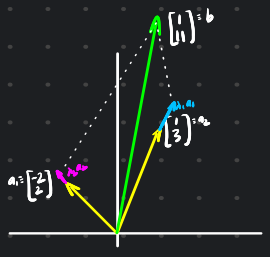
Linear combination of vectors

A linear combination of a list of vectors $v_1, \ldots, v_m$ in V is a vector of the form $a_1v_1 + \ldots + a_mv_m$, where $a_1, \ldots, a_m \in F$.

Example: linear combinⁿ in $\mathbb{R}^2$.

1.1. $(1, 11)$ is a linear combination of $(-2, 2)$ and $(1, 3)$.

$$1 \cdot (-2, 2) + 3(1, 3) = (1, 11)$$



Q. can we create $(0, 0)$ using $(-2, 2)$ and $(1, 3)$?

→ $a_1, a_2 \in \mathbb{R}$.

$$a_1 \begin{bmatrix} -2 \\ 2 \end{bmatrix} + a_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} = 0$$

$$\Rightarrow \quad \begin{aligned} -2a_1 + a_2 &= 0 \\ 2a_1 + 3a_2 &= 0 \end{aligned}$$

$$a_2 = 0$$

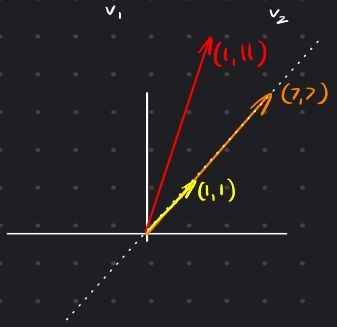
$$a_1 = 0$$

Q. can we create any 2 vectors in $\mathbb{R}^2$ using $(-2, 2)$ and $(1, 3)$?

→ yes. They span $\mathbb{R}^2$.

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consider $(1, 1)$ and $(7, 7)$. can we get $(1, 11)$ using v_1 and v_2?



NO.

→ they are restricted to this line.

can we get $(0, 0)$?

→ $a = 1, \; b = -\frac{1}{7}$

$$(1, 1) + \left(-\frac{1}{7}\right)(7, 7) = (0, 0).$$

here we get $(0, 0)$ without $a_1 = a_2 = 0$.

can we get any $v \in \mathbb{R}^2$?

→ NO.

Span of a list of vectors

It is the set of all linear combinations of the list of vectors.

$$\text{span}(v_1, \ldots, v_m) = \{a_1 v_1 + a_2 v_2 + \ldots + a_m v_m : a_1, \ldots, a_m \in \mathbb{R}\}$$

Lemma: Span(v_1, ..., v_m) is the smallest subspace containing v_1, ..., v_m.

(Any subspace contains the linear combinations (closure), thus containing the span as well).

Example: A list of vectors that span $\mathbb{F}^n / \mathbb{R}^n$:

Assume $n > 0$

$\{(1,0,\ldots,0), (0,1,0,\ldots,0), \ldots, (0,0,\ldots,0,1)\}$ → kronicher vectors

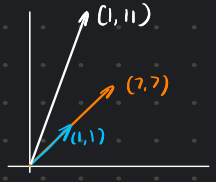
k^{th} vector is $(0,0,\ldots,0,1,0,\ldots,0)$ ↳ 1 in k^{th} position, rest 0.

always spans $\mathbb{F}^n$.

Proof: $x \in \mathbb{F}$ $\quad u_i = (0,0,\ldots,1,\ldots,0)$ ← 1 in i^{th} position.

$x = (x_1, \ldots, x_n) = x_1 u_1 + x_2 u_2 + \ldots + x_n v_n$.

Why were $(-2,2)$ and $(1,3)$ able to span $\mathbb{R}^2$, but $(1,1)$ and $(7,7)$ were not?



$(1,1)$ and $(7,7)$ are limited to a line.
We don't have enough information.
They are like a single vector in terms of information.
$(7,7)$ is in the span of $(1,1)$.
so $(7,7)$ doesn't give any additional information over $(1,1)$.

$(1,1)$ and $(7,7)$ are linearly dependent.

Linear independence

Suppose $v_1, \ldots, v_m \in V$ and $v \in \text{span}(v_1, \ldots, v_m)$.
By definition, $\exists a_1, a_2, \ldots, a_m \in \mathbb{F}$ s.t.

$$v = a_1 v_1 + a_2 v_2 + \ldots + a_m v_m. \quad \text{—①}$$

Is

Suppose $c_1, \ldots, c_m$ is another set of scalars s.t.

$$v = c_1 v_1 + c_2 v_2 + \ldots + c_m v_m. \quad \text{—②}$$

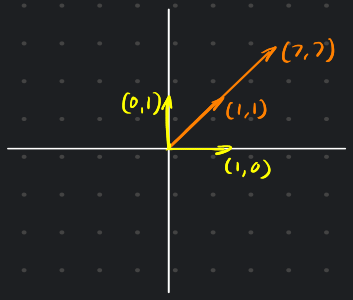
① - ②,

$$v - v = (a_1 - c_1) v_1 + (a_2 - c_2) v_2 + \ldots + (a_m - c_m) v_m$$
$$0 = (a_1 - c_1) v_1 + \ldots + (a_m - c_m) v_m$$

We can get $(0,0,\ldots,0)$ as a linear combination of $v_1, \ldots, v_m$.

If the only way to do this is by using (0) as the scalars, i.e. $a_i = c_i$,
then $v_1, \ldots, v_m$ are linearly independent.

LI: If $a_1 = a_2 = \dots = a_m = 0$ is the only solution to $a_1v_1 + \dots + a_mv_m = 0$, then $v_1, \dots, v_m$ are LI.



(0,1) and (1,0) give separate information.
We cannot get (1,0) from (0,1) and vice versa.
They are LI.

Alternate definition:
If any vector v is uniquely defined by the linear combination of $v_1, \dots, v_m$, then they are linearly independent.

Example 1: $\{(1,1)\} \rightarrow$ LI

Any list of of a single vector is LI, except if it is the zero vector.

Example 2: A list of length 2 in a vector space is LI iff neither of the 2 vectors are scalar multiples of the other.

Example 3: Generalization of the above.

If some vector in a list of vectors is a linear combination of the others, then the list is LD.

Proof: $v_m \in \text{span}(v_1, \dots, v_{m-1})$

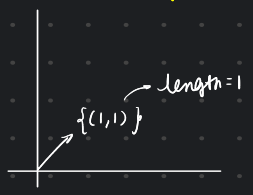
$\Rightarrow v_m = a_1v_1 + \dots + a_{m-1}v_{m-1}$

$\Rightarrow a_1v_1 + \dots + a_{m-1}v_{m-1} + (-1)v_m = 0$

$\therefore$ There is a soln other than $a_1 = a_2 = \dots = a_{m-1} = a_m = 0$.

$\Rightarrow$ They are LD.

Lemma: The length of a LI list $\leq$ length of a spanning list. (spans V)



A spanning list in $\mathbb{R}^2$ would have 2 vectors. (↳ minimum # of vectors)

length of the Kronecker vector (2 for $\mathbb{R}^2$) = length of a spanning vector.

Any 3 vector list $\{(1,0), (-1,-1), (-2,2)\}$

(these span $\mathbb{R}^2$) — so these cannot be LI.

Basis

A list that is LI and spanning V is a basis of V.

Dimension

The length of the basis of a vector space V is the dimension of V (length of minimum spanning list).

E.g. The standard basis of $\mathbb{R}^n$: $\{(1,0,0,\cdots,0), (0,1,0,\ldots,0), \cdots, (0,0,\ldots,0,1)\}$. Dim of $\mathbb{R}^n = n$.

Ex.2 : A basis of $\mathbb{R}^2$: $\{(-2,2)$ and $(1,3)\}$. Dim $(\mathbb{R}^2) = 2$.

Ex.3 : $\{(1,2,-4), (7,-5,6)\}$ is LI in $\mathbb{R}^3$ but does not span $\mathbb{R}^3$.

Ex.4. Consider the vector space $V = \{(x,x,y) \in \mathbb{R}^3 : x,y \in \mathbb{R}\}$
$\{(1,1,0),(0,0,1)\}$ is a basis of V.
Dimension of the space = 2.

Q. Which option is correct given $v_1,\ldots,v_4 \in \mathbb{R}^4$?

a) If $\{v_1,v_2,v_3\}$ is LD, so is $\{v_1,v_2\}$.

b) If v_4 is not a linear combination of $\{v_1,v_2,v_3\}$, then $\{v_1,v_2,v_3,v_4\}$ is LI.

c) Any set with the zero vector is LD.

d) If $\{v_1,v_2,v_3\}$ are LD, then so is $\{v_1,v_2,v_3,v_4\}$.

Q. We know that $a_1v_1+\cdots+a_nv_n=0$. Check if there's atleast one $a_i \neq 0$, then what ?
→ If yes, LI. If no, LD.

Q. If there's a set of vectors $v_1,\ldots,v_n$ s.t. $a_1v_1+\ldots+a_nv_n=0$ where $a_1=a_2=\cdots=a_n=0$,
is $\{v_1,\ldots,v_n\}$ LI?
→ False.

Q. $\{(4,4), (-1,3), (2,5), (8,3)\}$. check if LI.

→ length of any LI list ≤ length of spanning list. LD

length of basis of $\mathbb{R}^2 = 2$.

∴ The list cannot be LI.

Q. $\{(-8,-12,-4), (2,-\frac{3}{2},-1)\}$

→ Any list of vectors is LD iff one of them is a scalar multiple of the other. LI

so the list is LI as neither of them can be written as such.

Subspaces

H.W 2 (1) Check if $U = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 \cdot x_2 \cdot x_3 = 0\}$ is a vector space.

To check

(i) Closed under addition.

Let $x, y \in U$. then $x_1 \cdot x_2 \cdot x_3 = 0$

$y_1 \cdot y_2 \cdot y_3 = 0$

We want to check if $(x_1+y_1)(x_2+y_2)\cdot(x_3+y_3) = 0$

Subspaces

H.W 2 (1) Check if $U = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 \cdot x_2 \cdot x_3 = 0\}$ is a subspace. ✗ not a subspace

To check

(i) Closed under addition.

Let $x, y \in U$. then $x_1 \cdot x_2 \cdot x_3 = 0$

$y_1 \cdot y_2 \cdot y_3 = 0$

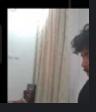
don't to be closed under addn.

We want to check if $(x_1+y_1)(x_2+y_2)\cdot(x_3+y_3) = 0$

Let $x = (1,1,0)$ & $y = (0,0,1)$

$x, y \in U$, $x+y \Rightarrow (1+0)\cdot(1+0)\cdot(0+1)$

$\neq 0$



LS

H.W 3 (B) $\{(4,4), (-1,3), (2,5), (8,1)\}$

Check if LS?

Linear maps

A linear map from V to W is a function T: V→W with the following properties:
i. Additivity: T(u+v) = Tu + Tv ∀ u, v ∈ V.
ii. Homogenity: T(λv) = λT(v) ∀ λ ∈ F, v ∈ V.

$T: V \to W$

$T(u+v) = T(u) + T(v)$

$T(\lambda v) = \lambda T(v)$

Notation

- Set of linear maps from V to W: $L(V, W)$
- Set of linear maps from V to V: $L(V)$

NOTE: $L(V, W)$ is a functional vector space.

Examples:

1. Identity operator: let $I \in L(V)$ s.t. $Iv = v \quad \forall\, v \in V$

$$I(u+v) = u+v = I(u) + I(v)$$
$$I(\lambda v) = \lambda v = \lambda \cdot Iv$$

2. let $T \in L(\mathbb{R}^3, \mathbb{R}^2)$ s.t.

$$T(x, y, z) = (2x - y + 3z,\ 7x + 5y - 6z)$$

3. In general from $\mathbb{F}^n$ to $\mathbb{F}^m$ for $n, m > 0$, let $A_{j,k} \in \mathbb{F}$ for each $j = 1, \dots, m$ and $k = 1, \dots, n$

and $T \in L(\mathbb{F}^n, \mathbb{F}^m)$ s.t.

$$T(x) = T(x_1, x_2, \dots, x_n)$$
$$= (A_{11}x_1 + \dots + A_{1n}x_n,\ \dots,\ A_{m1}x_1 + \dots + A_{mn}x_n)$$

$$\begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \vdots & & & \\ A_{m1} & A_{m2} & \cdots & A_{mn} \end{bmatrix}_{m \times n} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1} = \begin{bmatrix} A_{11}x_1 + \dots + A_{1n}x_n \\ \vdots \\ A_{m1}x_1 + \dots + A_{mn}x_n \end{bmatrix}_{m \times 1}$$

Matrices are linear maps.

$$AX = T(x)$$

Any linear map $\mathbb{R}^m \to \mathbb{R}^n$ can be represented in matrix form as above.

$T(0) = 0$ for every linear map T.

Proof: T is a linear map. so it follows additivity.

$$T(0+0) = T(0) + T(0)$$
$$\Rightarrow T0 = T0 + T0$$
$$\Rightarrow T0 = 0$$

Null spaces

For $T \in L(V, W)$,

$\text{null } T = \{ v \in V : Tv = 0 \}$

The null space of $T: V \to W$ is the subset of V of those vectors that are mapped to 0 by T.

e.g. If T is zero mapped from V–W, i.e. $Tv = 0 \quad \forall \ v \in V$.

$\text{null } T = V$ → subspace of V.

2. $T \in L(\mathbb{R}^3, \mathbb{R}^2)$ s.t. $T(x,y,z) = (2x - y + 3z, 7x + 5y - 6z)$

$$\text{null } T = \left\{ x, y, z \in \mathbb{R}^3 : \begin{matrix} 2x - y + 3z = 0 \\ 7x + 5y - 6z = 0 \end{matrix} \right\}$$

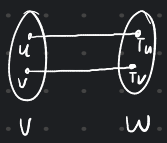
↳ subspace of V.

null T is a subspace of V (domain of T).

Injectivity (one-to-one)

A function $T: V \to W$ is injective if

$Tu = Tv$ implies $u = v$.



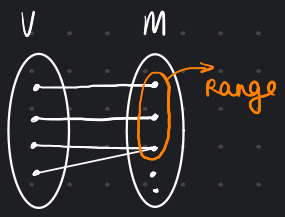
Let $T \in L(V, W)$. Then T is injective iff $\text{null } T = \{0\}$.

↳ columns of T are LI.

Range of T

For $T \in L(V, W)$, the range of T is the subset of W (→ subspace of W) containing those vectors that are equal to Tv for some $v \in V$.

$$\text{range } T = \{ Tv : v \in V \}$$



Range = All vectors in W that can be written as Tv for some $v \in V$.

$$\begin{bmatrix} | & & | \\ t_1 & \cdots & t_n \\ | & & | \end{bmatrix} \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$$

↳ matrix of T ↳ some $v \in V$.

$Tv \ \forall \ v \in V = \text{range}(T)$.

If we consider all the values of v possible, we get the span of T.

Surjectivity (onto)

A function $T: V \to W$ is surjective if

range T = W.

